

Chipmoji

An emoji guide for chip development commits.

GIT 16

	:tada:
Initial commit or project bootstrap.	
	:bug:
Fix functional bugs in RTL, verification, Python, scripts, or flow code.	
	:ambulance:
Fix urgent breakage: failing regression, broken CI, lab blocker, release blocker, or tapeout blocker.	
	:sparkles:
Add a complete new feature.	
	:construction:
Commit incomplete work intentionally.	
	:boom:
Breaking interface, register map, protocol, API, timing contract, or CLI change.	
	:recycle:
Refactor without intended behavior change.	
	:art:
Improve structure, formatting, naming, or readability without behavior change.	
	:fire:
Remove obsolete code, files, stale scripts, or generated artifacts.	
	:coffin:
Remove dead signals, unreachable FSM states, unused functions, or obsolete helpers.	
	:wastebasket:
Deprecate code or behavior that should be removed later.	
	:truck:
Move or rename files, modules, packages, paths, or resources.	
	:rewind:
Revert a previous change.	
	:twisted_rightwards_arrows:
Merge branches. Do not use for muxing, routing, or arbitration.	
	:see_no_evil:
Add or update <code>.gitignore</code> .	
	:bookmark:
Release or version tag.	

DOCS 4

	:memo:
Add or update docs, specs, register maps, integration notes, or README content.	
	:bulb:
Add or update source comments explaining hardware behavior, assumptions, or constraints.	
	:pencil2:
Fix typos in docs, comments, specs, or messages.	
	:page_facing_up:
Add or update license.	

TIMING 2

	:scissors:
Reduce combinational depth, retime logic, split paths, or improve Fmax.	
	:stopwatch:
Benchmark instrumentation, cycle counting, latency measurement, or timing measurement infrastructure.	

CI 2

	:construction_worker:
CI pipelines, GitHub Actions, build matrix, scheduled jobs, or regression automation.	
	:green_heart:
Fix broken CI, regression automation, build matrix, or tool-cache issue.	

RTL 12

	:bridge_at_night:
Clock-domain crossing, async boundary, synchronizer, handshake bridge, or reset-domain crossing.	
	:rightwards_pushing_hand:
Valid/ready pushback, stall propagation, skid buffers, flow control, or downstream backpressure.	
	:electric_plug:
Wire ports, buses, protocols, register interfaces, AXI/APB/TileLink/Wishbone connections, or top-level integration.	
	:abacus:
Arithmetic, bit-width, fixed-point, saturation, rounding, accumulator, multiplier, divider, or datapath logic.	
	:vertical_traffic_light:
Arbitration, priority, scheduling, traffic shaping, QoS, credits, or grant logic.	
	:compass:
Address maps, memory maps, topology, routing tables, NoC coordinates, or integration direction.	
	:dna:
Generics, parameters, SKUs, feature variants, configurable widths/depths, or generated RTL variants.	
	:fire_extinguisher:
Fix X-prop, glitch risk, unsafe reset behavior, metastability risk, race, or silicon-escape-class hazard.	
	:magnet:
Board-level work, IO, pins, PHYs, lab-only fixes, FPGA bring-up, or constraints tied to real hardware.	
	:satellite_antenna:
PCIe, Ethernet, DDR, SerDes, SPI, I2C, UART, or other external protocol/link behavior.	
	:safety_pin:
Small localized hardware workaround or mitigation for a known issue.	
	:screwdriver:
Small mechanical RTL adjustment too narrow for a bigger category.	

PYTHON 5

	:snake:
Python-specific implementation work when no more precise standard Gitmoji applies.	
	:label:
Add or update Python type hints, dataclasses, TypedDicts, Pydantic models, SV typedefs, enums, or structs.	
	:goal_net:
Improve exceptions, parser errors, failure handling, or user-facing error messages.	
	:monocle_face:
Data exploration, report inspection, one-off analysis, or investigation scripts.	
	:thread:
Threading, multiprocessing, async, job queues, or concurrent report processing.	

INFRASTRUCTURE 2

	:bricks:
Hardware-flow infrastructure: runners, artifact storage, caches, job orchestration, or shared tool setup.	
	:package:
Add or update generated bundles, packaged IP, release artifacts, reports, or bitstreams.	

VERIFICATION 7

	:white_check_mark:
Add, update, or fix passing tests, regressions, assertions, formal checks, or checkers.	
	:test_tube:
Add a failing test or reproducer for a known bug.	
	:rotating_light:
Fix compiler, linter, CDC, synthesis, simulation, or timing warnings.	
	:game_die:
Randomized testing, constrained-random stimulus, fuzzing, seeds, or randomized regressions.	
	:microscope:
Add probes, traces, counters, waveform hooks, debug CSRs, or observability.	
	:safety_vest:
Add validation, guards, illegal-state checks, protocol checks, or defensive checks.	
	:seedling:
Add seed files, sample inputs, golden data, or deterministic regression data.	

DEPENDENCIES 6

	:arrow_up:
Upgrade dependency, vendor IP, tool version, package, or submodule.	
	:arrow_down:
Downgrade dependency, vendor IP, tool version, package, or submodule.	
	:pushpin:
Pin exact versions of tools, IPs, packages, lockfiles, containers, or submodules.	
	:heavy_plus_sign:
Add a dependency, package, included IP, submodule, or tool requirement.	
	:heavy_minus_sign:
Remove a dependency, package, included IP, submodule, or tool requirement.	
	:alien:
Update code due to external API, protocol, vendor IP, tool output, or report-format changes.	

PPA 4

	:zap:
Improve speed, throughput, latency, benchmark result, or runtime.	
	:chart_with_downwards_trend:
Reduce LUTs, FFs, gates, SRAM, BRAM, DSPs, cells, or general resource usage.	
	:chart_with_upwards_trend:
Accept area, power, or resource increase intentionally for feature, timing, verification, or performance.	
	:battery:
Clock gating, power gating, idle behavior, low-power mode, activity reduction, or power optimization.	

SCRIPTS 2

	:hammer:
Build, sim, synth, lint, formal, regression, packaging, or helper scripts.	
	:wrench:
Tool config, environment config, CLI config, YAML/TOML/JSON config, synthesis/sim flags, or parameters.	